Perceptual vs. Objective Evaluation of ML Music Source Separation on Electronic Music

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INTRODUCTION/MOTIVATION

Music source separation (MSS) - the process of extracting individual stems from a mixed recording – has a variety of applications in remixing, production, and interactive media. ML-based approaches to MSS have advanced rapidly become the dominant solution to this problem, but evaluation of these approaches rely heavily on objective metrics (SDR, FAD) that may not reflect human perception. This is a particularly interesting point for non-traditional genres like electronic music, where heavy processing complicates both separation and evaluation. This study compares objective and subjective evaluations of three representative commercial MSS models on electronic drum and bass stems.

BACKGROUND & OBJECTIVES

Three model architectures were evaluated:

- **Spleeter (Deezer):** Solely spectrogram-domain, struggles with phase artifacts
- **Demucs v2:** Older version of Demucs, solely waveform-domain; raw audio with a bidirectional LSTM
- **HT-Demucs:** Hybrid domain, utilizes transformer architecture

Objective evaluation metrics included SDR (Signal-Distortion Ratio) and FAD (Frechet Audio Distance). Prior work (Rumbold et al., 2024; Cano et al., 2016) found no significant correlation between these objective metrics and human listening preference, motivating the need for tightly controlled perceptual studies, especially on underrepresented genres.

METHODS

Stimuli: 6 electronic tracks from the MUSDB18 and Native Instruments STEMS datasets. Drum and bass stems were separated by each model in the typical 4-stem configuration. Vocal and instrumental stems were discarded due to variation in content and lack of clarity in the ground truth stems. Anchor stems were created by low-passing the reference at 3.5 kHz and applying bit-depth reduction, in according with common practice in the literature.

Participants: 15 Music Engineering grad/undergrad students at the University of Miami, selected for familiarity and adeptness at critiquing audio stimuli

Procedure: MUSHRA-style online questionnaire (Qualtrics). Full mix provided as listening context in place of isolated reference stem to reflect real-world usage conditions (Wierstorf et al., 2017). Participants rated each stimulus 0-100 (dreadful → perfect). Order of trials and stimuli were randomized to prevent fatigue/order effects. Objective scores (SDR, FAD) were calculated using MIR Eval and FAD libraries in Python

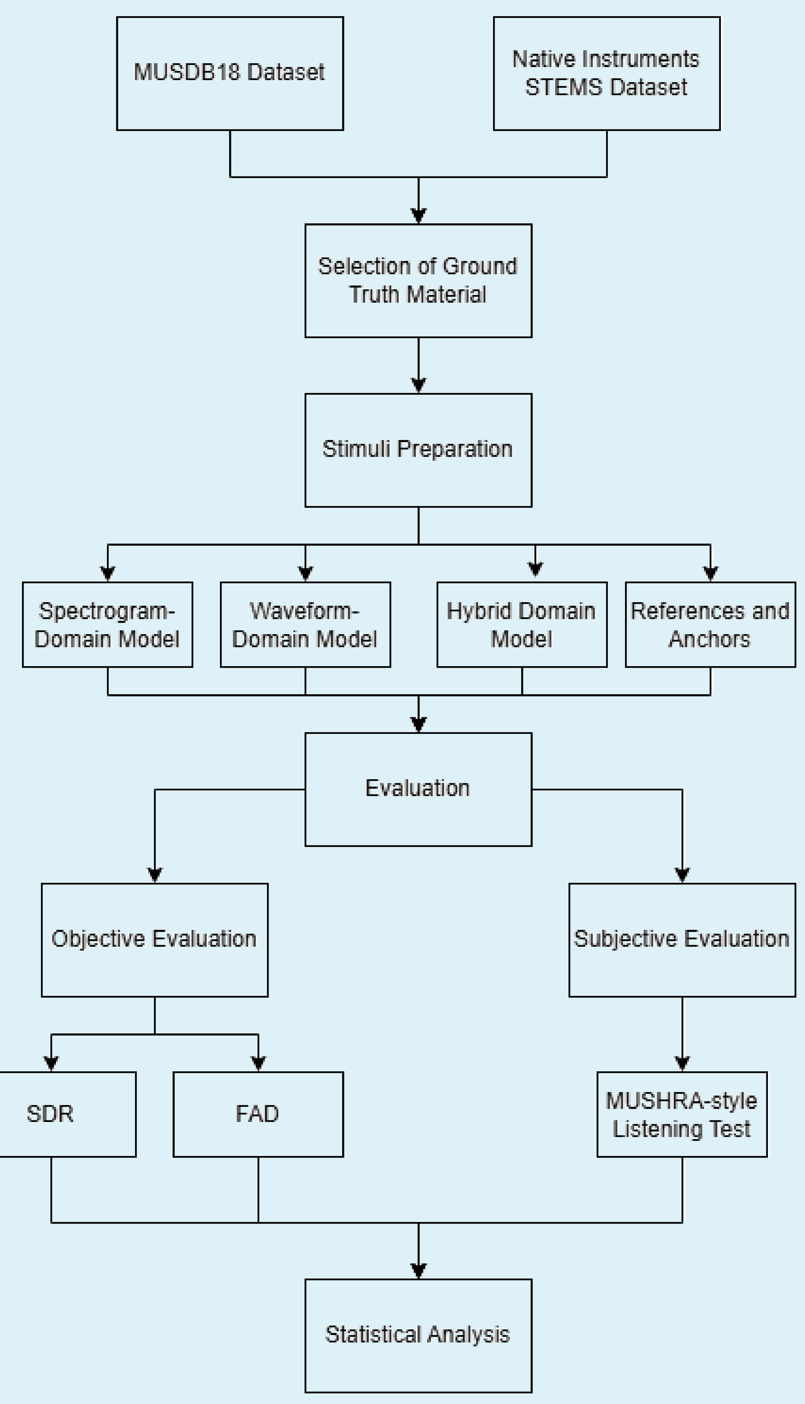

Figure – Experimental flow chart

RESULTS (OBJECTIVE)

SDR and FAD agree consistently across multiple embedding spaces (VGGish, PANN). HTDemucs outperformed both single-domain models across all conditions. ANOVA confirmed significant main effects of model (p = .032) and stem type (p = .04), with bass stems yielding higher SDR than drums.

Model	Avg. SDR (Bass)	Avg. SDR (Drums)
anchor	-14.07	-11.78
dv2	7.29	2.29
htdemucs	14.83	5.30
reference	226.84	239.60
spleeter	2.35	1.81

Table 1 – SDR Results

model	FAD (pann)	FAD (vggish)
anchor	93.68	14.05
dv2	39.84	4.05
htdemucs	30.26	3.72
reference	0.00	0.00
spleeter	47.24	3.81

Table 2 – FAD Results

RESULTS (SUBJECTIVE)

12 of 15 participant responses were admitted. MUSHRA ratings were normalized as z-scores to account for variation in range of responses. A t-test showed HTDemucs rated statistically equivalent to the ground-truth reference stem ($t(71) = -0.514, p = 0.6092$), a noticeable result. Single-domain models rated significantly lower. A significant interaction between model and stem type is observed, where single-domain models were perceived as performing much worse than expected despite the objective scores showing higher SDR ratings, indicating that the transient nature of drum stems may mask separation artifacts.

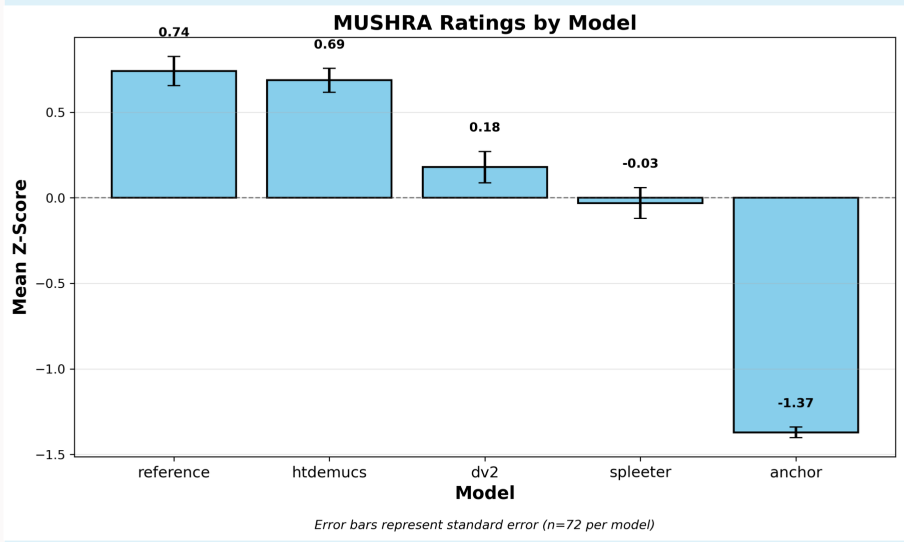

Figure – Normalized MUSHRA ratings by model

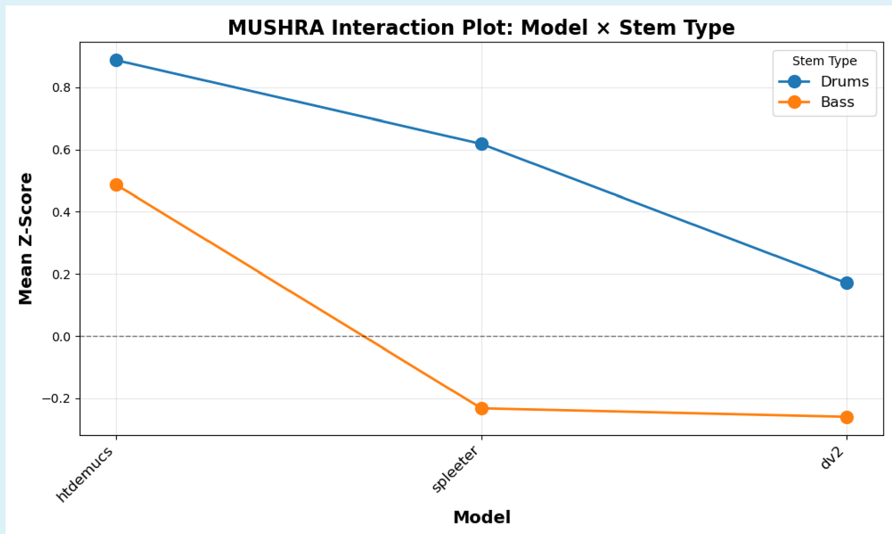

Figure – MUSHRA ANOVA interaction plot

CORRELATION ANALYSIS

Spearman rank correlation was done between trial-level MUSHRA scores and SDR; $\rho = 0.459, p = 0.0552$ ($n = 18$), indicating moderate correlation but falling short of the significance threshold. Unlike prior contributions (Rumbold et al., 2024), which found no significant correlation, our moderate rho-value suggests that SDR can capture overall model differences but lacks the resolution to capture finer perceptual distinctions, most likely due to its equal weighting of all distortions regardless of frequency/audibility.

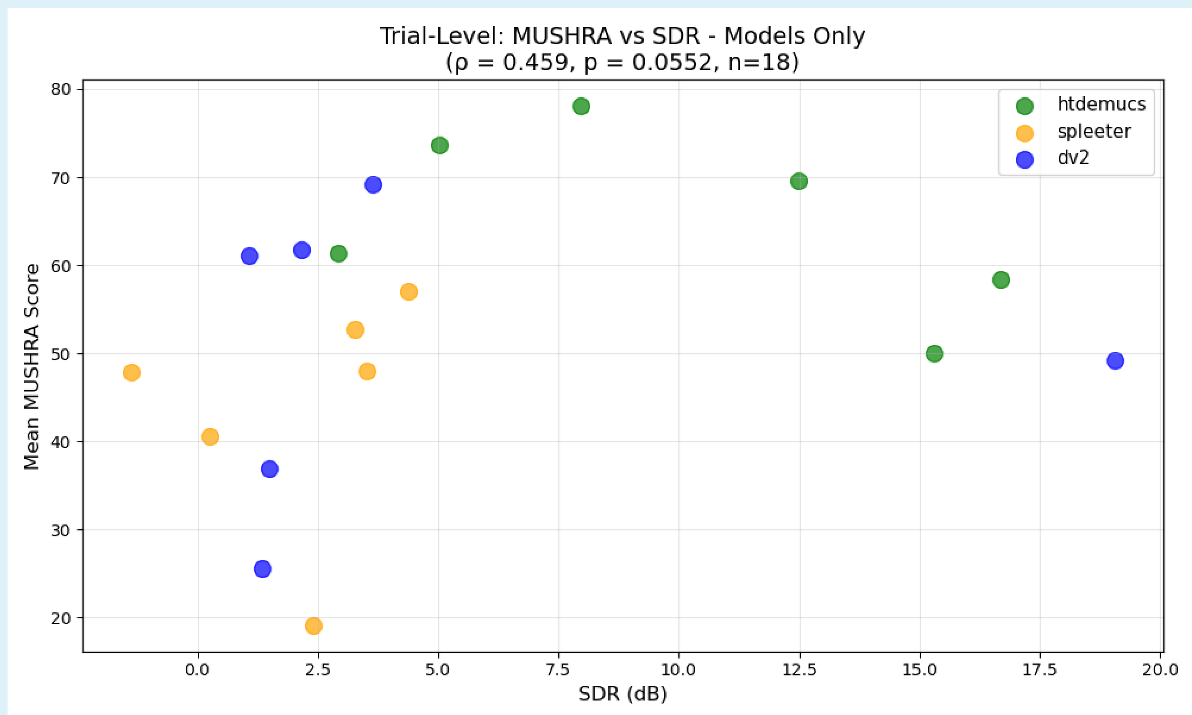

Figure 3 – Averaged MUSHRA scores plotted against SDR for each trial (n=18)

CONCLUSIONS & FUTURE WORK

HTDemucs consistently outperformed single-domain models across all metrics. Perceptual and objective evaluations agreed broadly but diverged on stem-type differences, highlighting the limitations of SDR. The perceptual equivalence of HTDemucs output to the ground truth reference stems may reflect genre and use-case specific listener expectations shaped by the heavily processed nature of electronic music.

Future work of a similar vein should seek to include:

- Larger sets of genre-specific stimuli
- Broader participant pools,
- Use of perceptually weighted metrics (PEASS) for comparison
- Inclusion of vocal/instrumental stems to represent more use cases

This study reaffirms the importance of rigorous perceptual evaluation to MSS model development, rather than reliance on purely computational metrics.